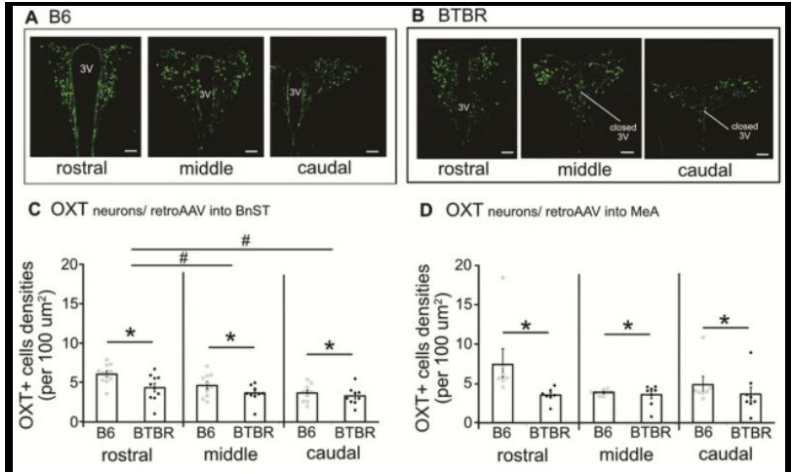
# Oxytocin neurons in the paraventricular nucleus and fear empathy among male mice

Lu Zhang, MS et al.

Presented by: Emelia Delgado, Raelyn Mornay and Cass Vandevoorde

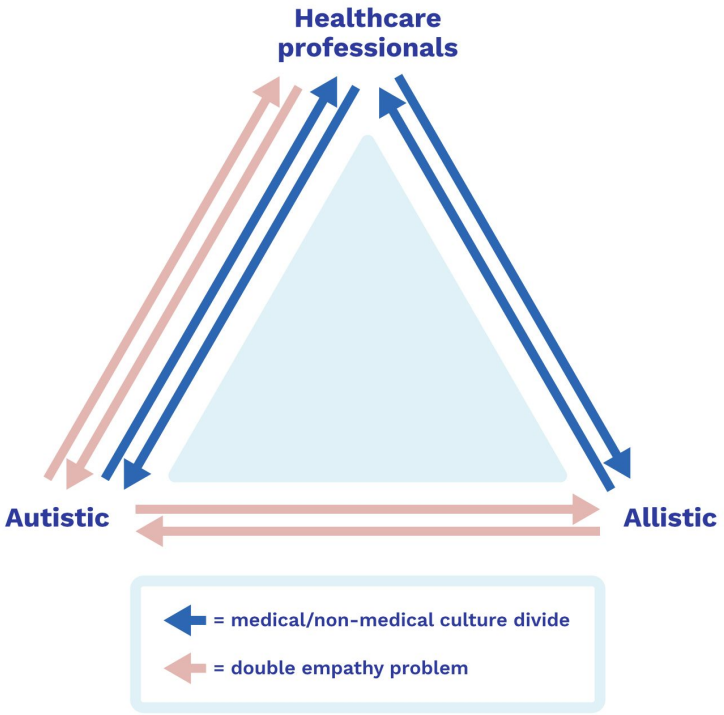
# Paraventricular nucleus and its role in oxytocin pathways




Arakawa et al., 2023

- Previous studies found oxytocin cell densities were significantly lower in the paraventricular nucleus of mice modeling autism
- High concentration of OXT neurons are found in the PVN which connects it to other oxytocin circuits
- OXT circuits were shown to regulate social behaviors

# Empathy in ASD




- Reduced empathy has been correlated with ASD
- The Double Empathy problem (Milton 2012) and The Triple Empathy Problem (Shaw et al., 2023) → Oppose Baron-Cohen's Theory of Mind
- Backed by studies that show strong peer communication between autistic individuals (Crompton et al., 2020)
- Autism has a 4:1 male to female diagnosis ratio, male mice are used

# Hypothesis

The researchers sought to compare brain regions associated with empathy-like behaviours of fear and pain. They explored the role of the oxytocin–oxytocin receptor system in fear empathy.

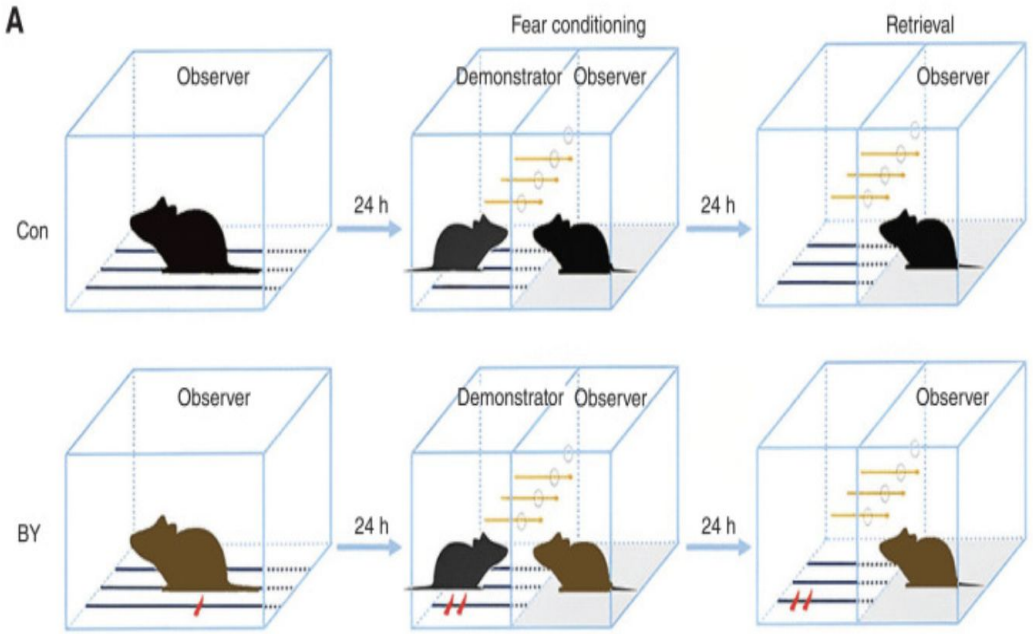
# Importance

They explored these factors to gain a deeper understanding of the neural mechanisms underlying empathy-like behaviours of fear and pain, shedding light on the intricate nature of empathic responses.

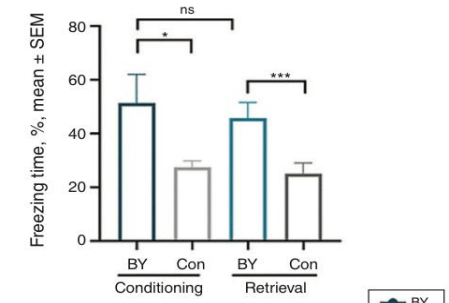
# Fear Empathy Model

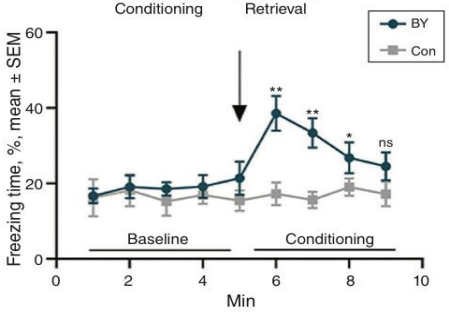
**Purpose**: Analyze if the observer mice showed any fear from watching the demonstrator mice.

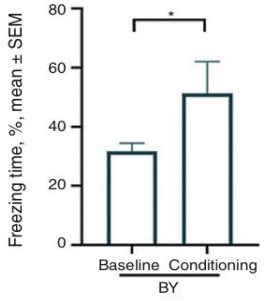
- The **observer** mouse watches **demonstrator** mice get shocked to replicate seeing someone in fear.
- The **demonstrator** mice received small electrical shocks on their feet.
- Both groups of **C57BL mice** observed the fear box 3 days before the experiment to become familiar with the environment

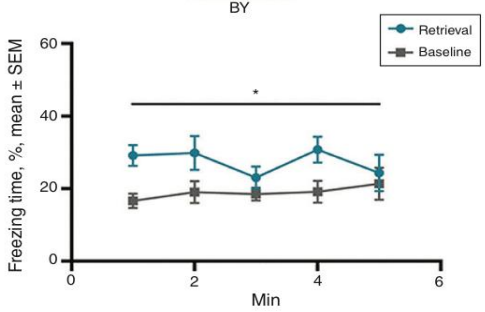


# Figure 1 : Oxytocin Neuron Manipulation



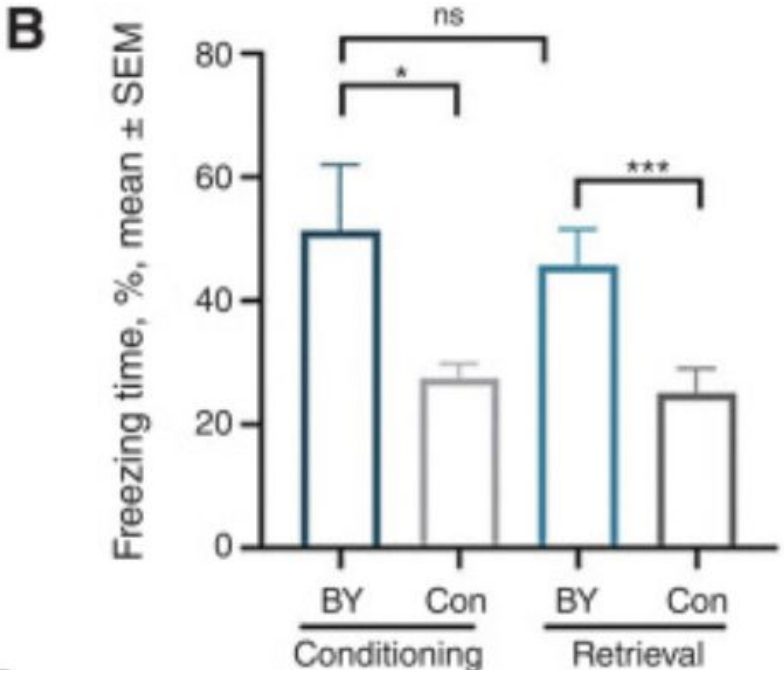




**Key Takeaway:** The mice show empathy like fear responses when they watch other mice in distress and have prior exposure to the fear box

- Mice showed empathy like fear just by watching other mice in distress.
- The mice remembered the fear box within a 24 hour time span

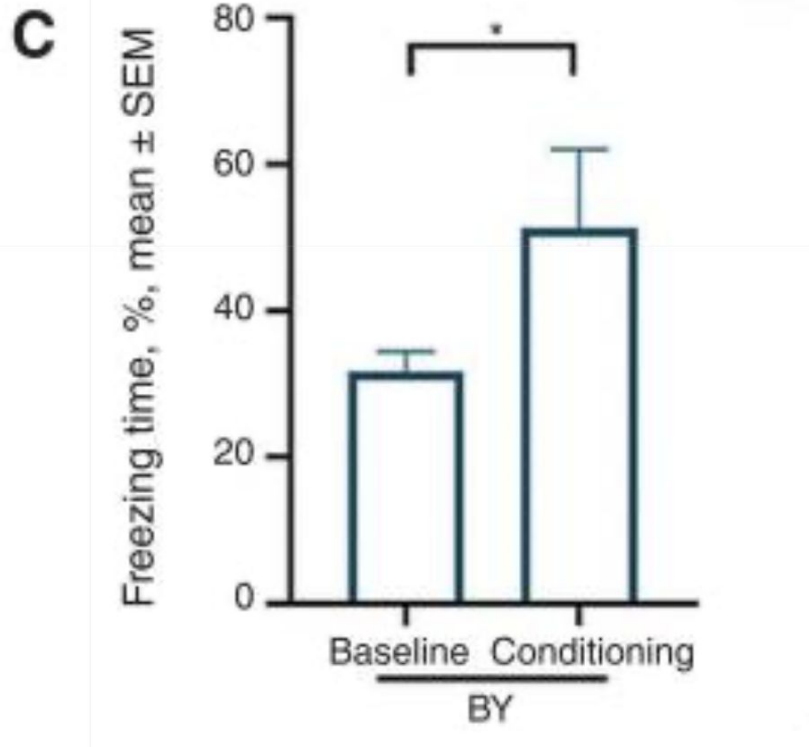
# Figure 1: Panel B Freezing Time




**Key Takeaway**: The mice in the bystander group learned fear by observing and showing empathy.

- The BY mice froze more than the control mice in both the retrieval and conditioning responses.
- Freezing % for the BY mice was high when they watched the control mice get shocked and before the electrical shock

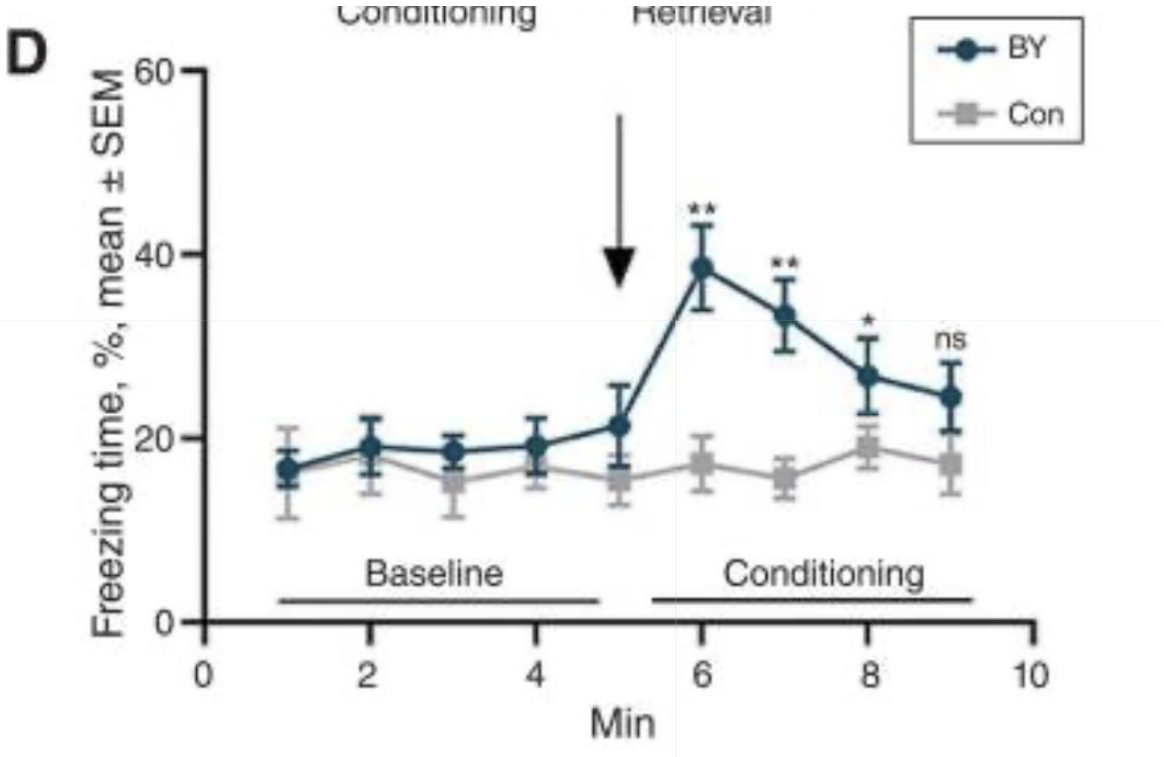
# Figure 1: Panel C




**Key Takeaway:** Watching another mice suffer triggered a fear response and reaction in both the bystander group and control group.

- When comparing the freezing levels between the BY mice before and after watching experimental mice there was a significant increase
- Comparing By mice freezing % before and after fear conditioning test

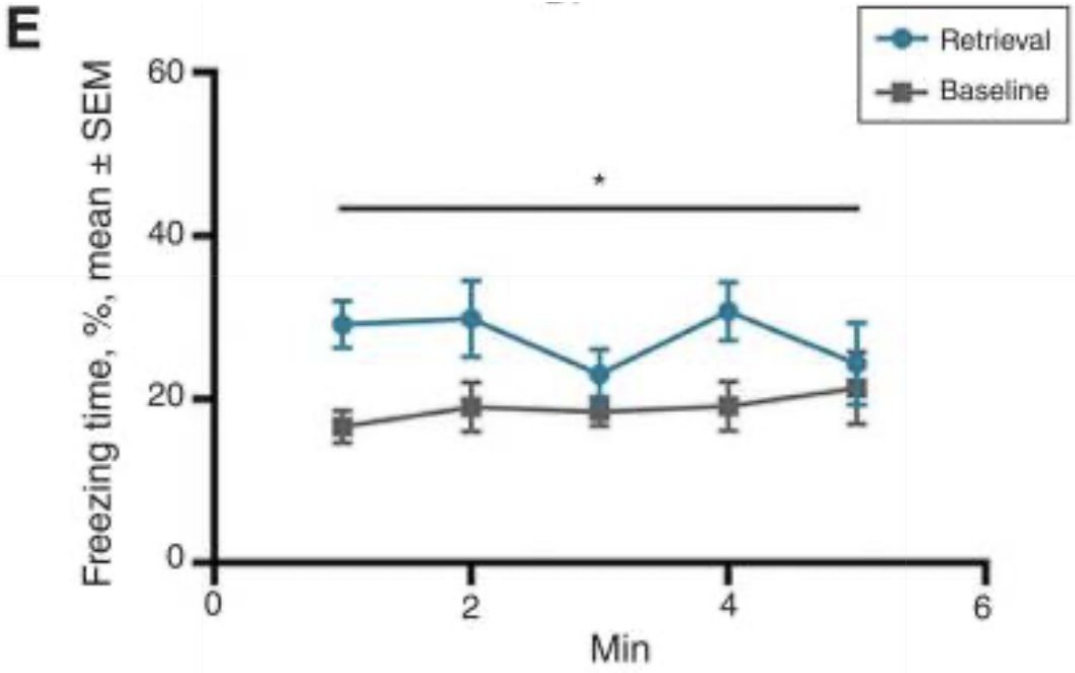
# Figure 1: Panel D Freezing Time After 24 hours




**Key Takeaway:** BY mice froze more when observing another mice suffering emotionally once the shocks started

- Freezing time over the span 2 days shows the control group stayed the same mostly and the bystander group immediately increased freezing once the shocks started after 5 minutes

# Figure 1: Panel E Day 3 Shock Retrieval Memory



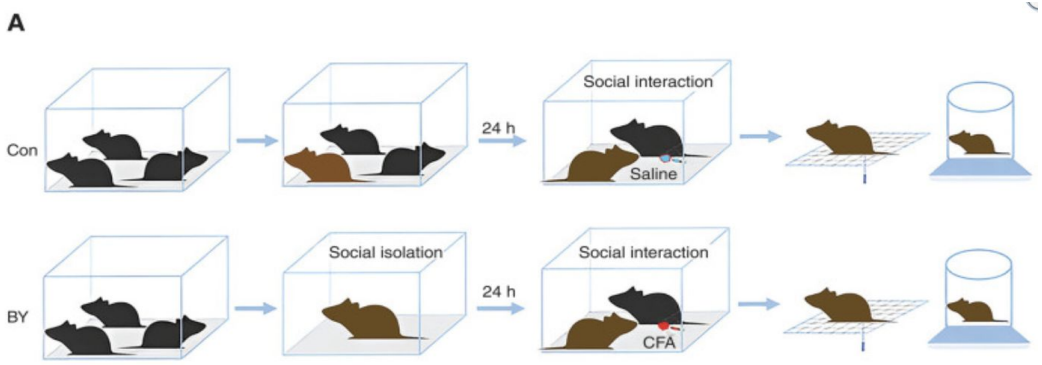

**Key Takeaway**: The mice were not only scared in the moment they remember the fear box but also showed increased signs of freezing

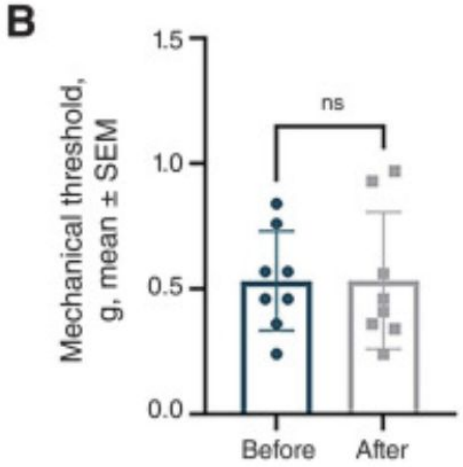
- The bystander mice froze more during the retrieval of the experimental mice getting shocked even when the demonstrator mice were not being shocked

# Figure 2: Pain Empathy Model

Purpose: Measuring how social isolation and social interaction with a mice in pain affects pain sensitivity

- BY group was injected with Freund's adjuvant 10 microliters in the left paw
- Data supports the hypothesis that social and emotional experiences can influence pain perception
- Pain sensitivity was measured after 24 hours of isolation and then after social interaction

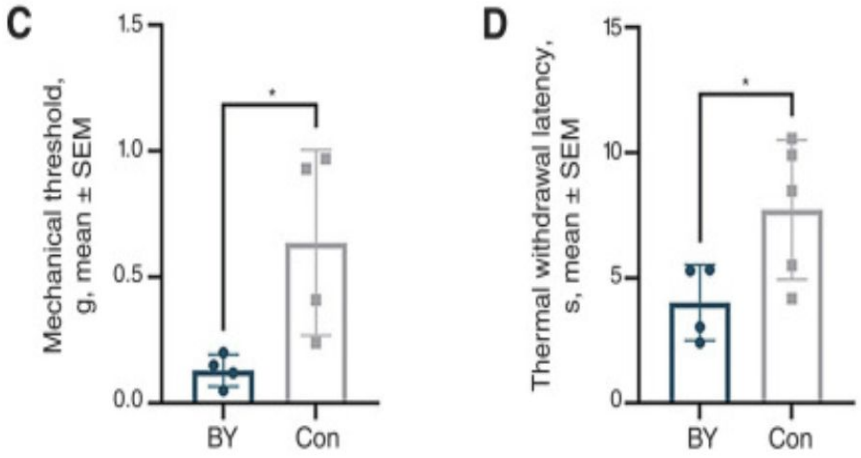



**Key Takeaway:** Being socially isolated did not have any significant difference on pain threshold

- Freund's adjuvant localized inflammation and pain

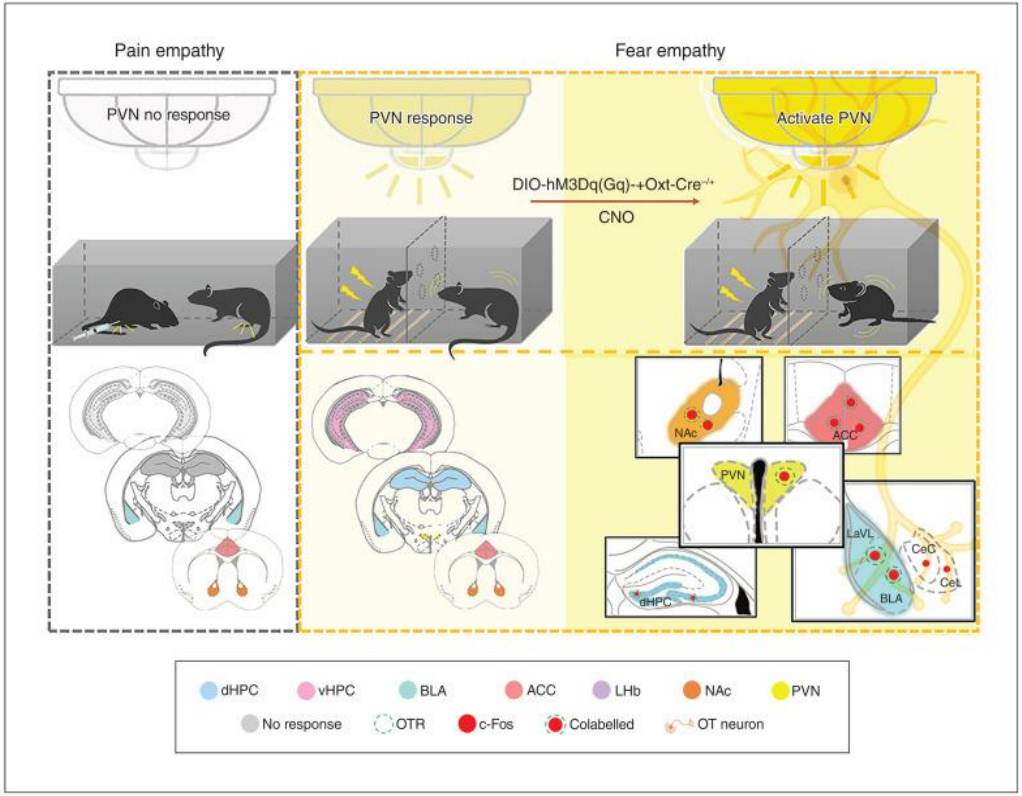
# Figure 2: Panel C & D Pain Sensitivity




**Key Takeaway:** After social interaction mechanical threshold and thermal withdrawal of observers in the BY group were significantly lower

- Social interaction with a peer in pain affects the mice pain perception
- BY mice showed increased pain sensitivity as the demonstrator mice suggesting pain - empathy increased
- Bystander mice are more sensitive to pain after being exposed to a demonstrator mice being in pain

# Figure 6: Brain regions of empathy-like behaviours of fear and pain in C57BL mice




**Key Takeaway**: Different brain regions are activated for Pain Empathy and Fear empathy

- PVN activity was seen in fear empathy over pain empathy
- Specific activation of oxytocin neurons in the PVN region is positively associated with empathy

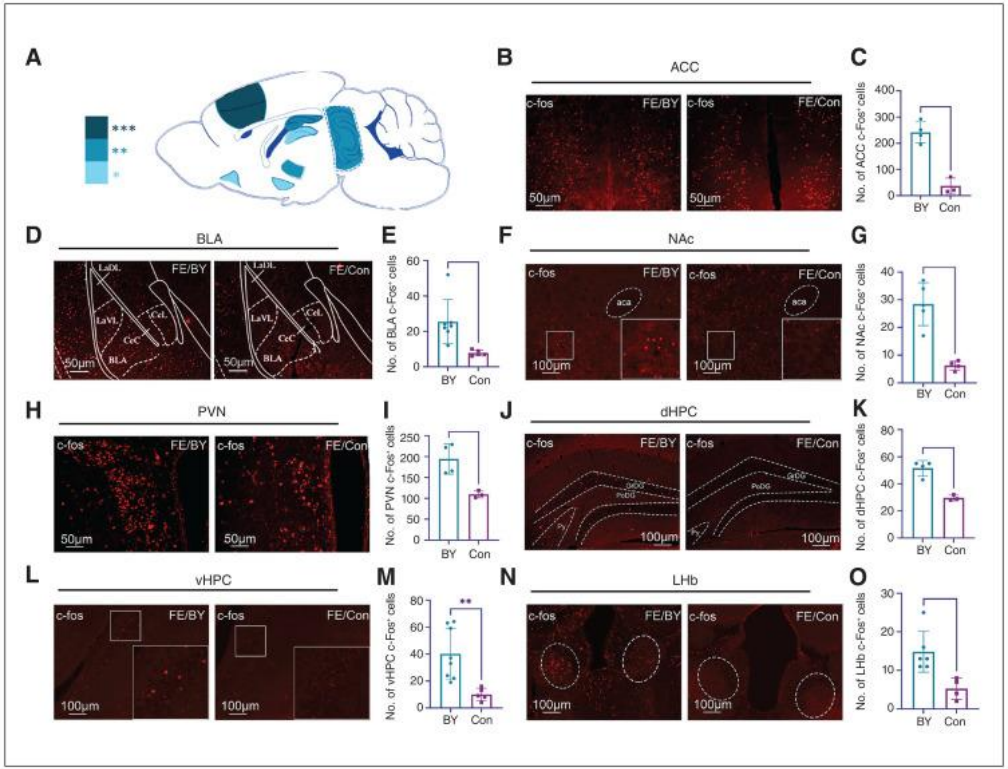
Abbreviation key:

- dHPC = dorsal hippocampus
- vHPN = ventral hippocampus
- BLA = basolateral amygdala
- ACC = anterior cingulate cortex
- LHb = lateral habenula
- NAc = nucleus accumbens
- PVN = paraventricular nucleus
- OTR = oxytocin receptor
- OT neuron = oxytocin neuron

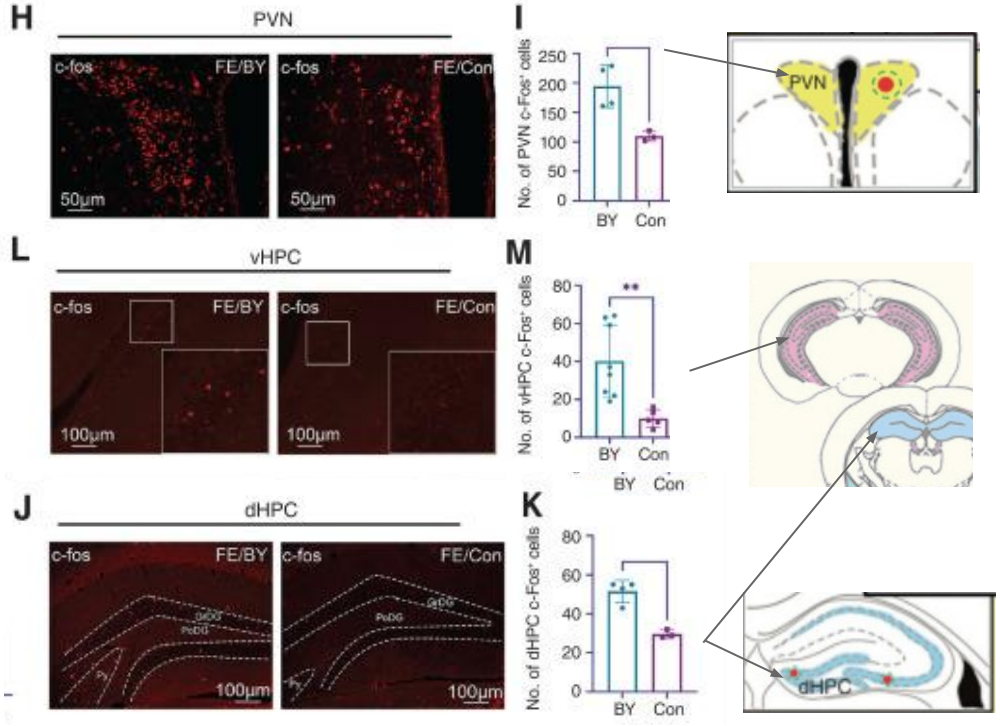
# Relevant functions of Brain regions associated with empathy-like behaviours and oxytocin pathways

| | | |
|---|---|---|
| dHPC | Dorsal Hippocampus | Spatial learning and memory |
| vHPC | Ventral Hippocampus | Emotional regulation, anxiety |
| BLA | Basolateral Amygdala | Connects stimuli and emotion |
| ACC | Anterior Cingulate Cortex | Decision making, pain perception |
| LHb | Lateral Habenula | Aversions, “anti-reward” system |
| NAc | Nucleus Accumbens | Motivation, reward circuit |
| PVN | Paraventricular Nucleus | Major oxytocin production cite |
| OTR | Oxytocin Receptor | Binds Oxytocin |
| OT neuron | Oxytocin Neuron | Synthesize and release Oxytocin |

# Figure 4: Brain staining and c-Fos–positive cell counts in brain regions involved in the fear empathy in C57BL mice



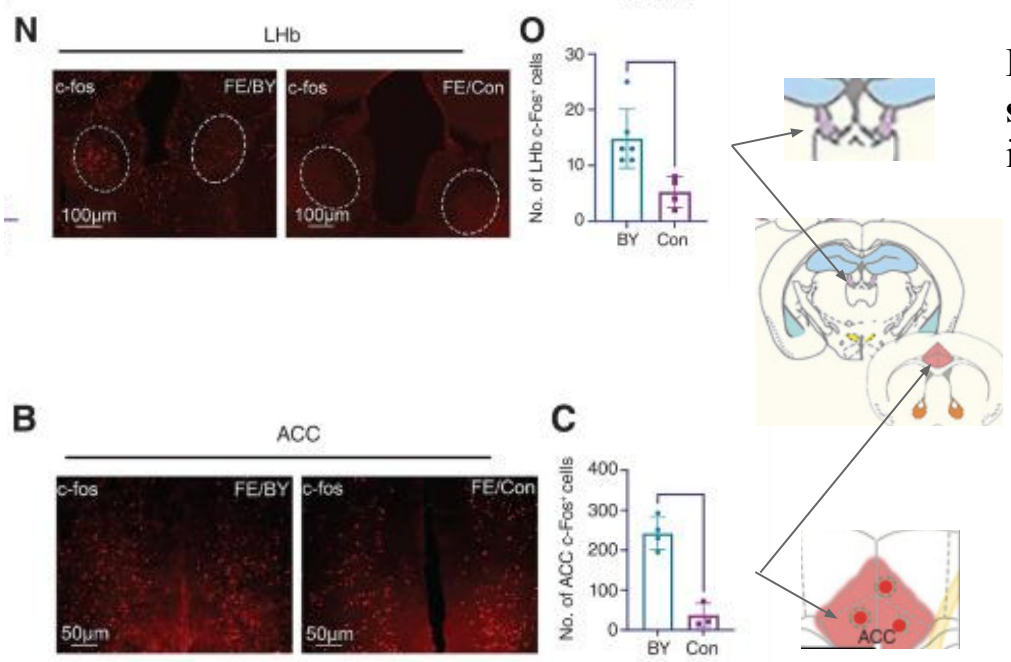
# Figure 4: Brain staining and c-Fos–positive cell counts in brain regions involved in the fear empathy in C57BL mice




**Key Takeaway:** the PVN, dorsal hippocampus, and ventral hippocampus were significantly activated in empathy-like fear responses, but not empathy-like pain responses

- **PVN** shows the most pronounced **significant** difference in c-Fos expression
- **Significantly** higher of c-Fos–positive cells were found in the BY groups
- Not seen in corresponding group in Figure 5

# Figure 4: Brain staining and c-Fos–positive cell counts in brain regions involved in the fear empathy in C57BL mice

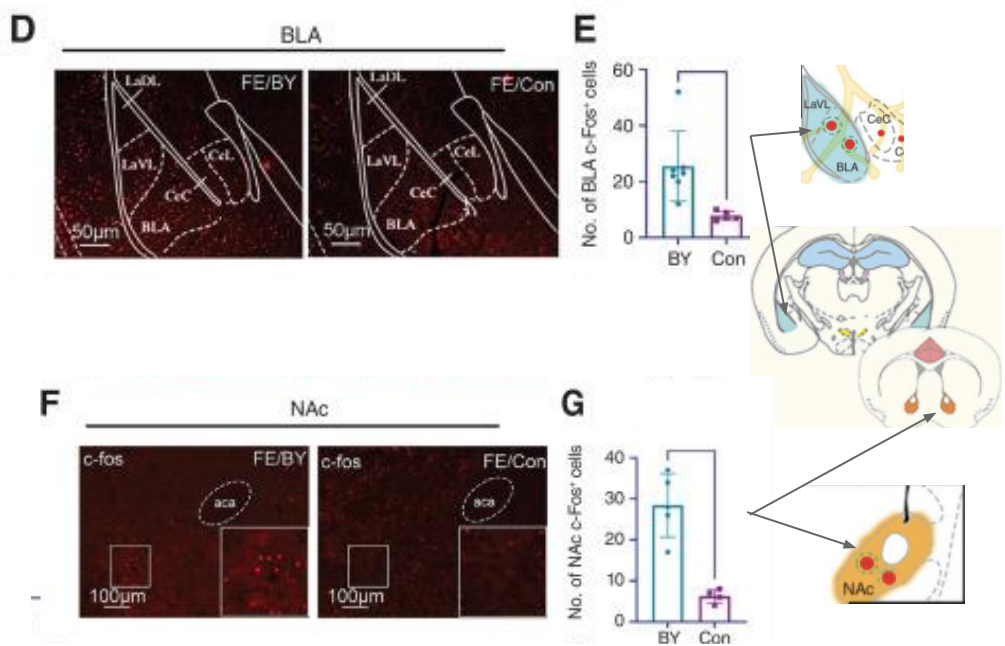



**Key Takeaway:** Both LHb and ACC showed a **significantly** higher number of cfos positive cells in the BY group

- LHb and ACC had **significant** activation for both types of empathy responses
- LHb showed stronger activation for **empathy-like response to pain**
- ACC showed stronger activation for **empathy-like response to fear**

Fig. 4

# Figure 4: Brain staining and c-Fos–positive cell counts in brain regions involved in the fear empathy in C57BL mice

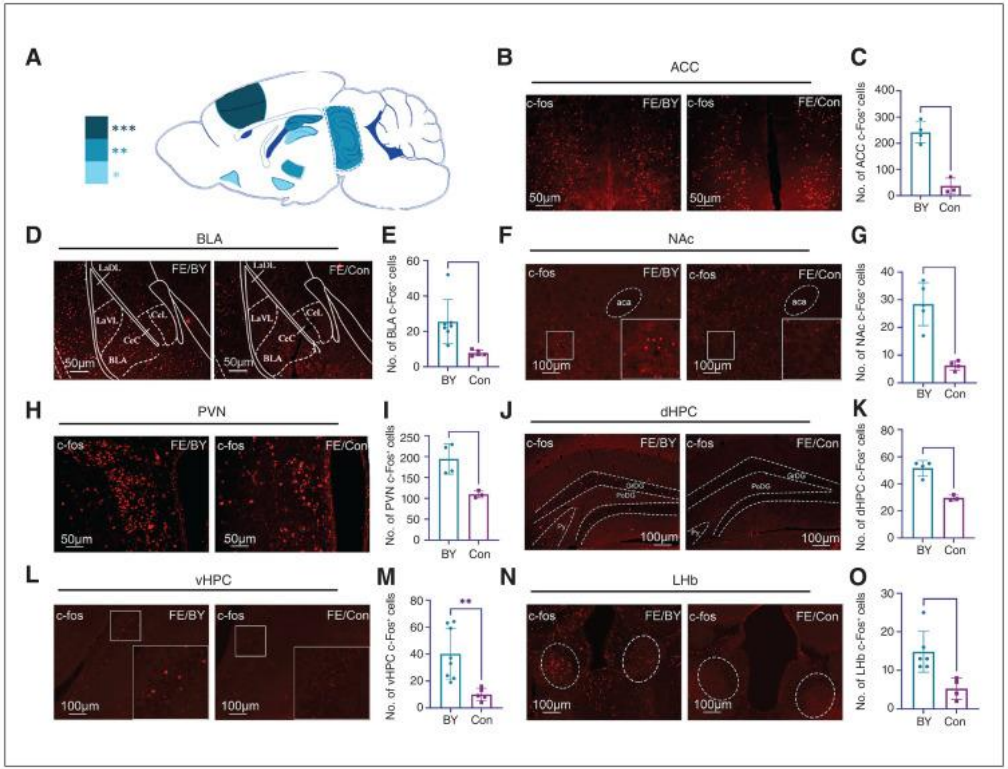



**Key Takeaway:** BLA and NAc showed a **significantly** higher number of cfos positive cells in the BY group for both types of empathy-like behaviours

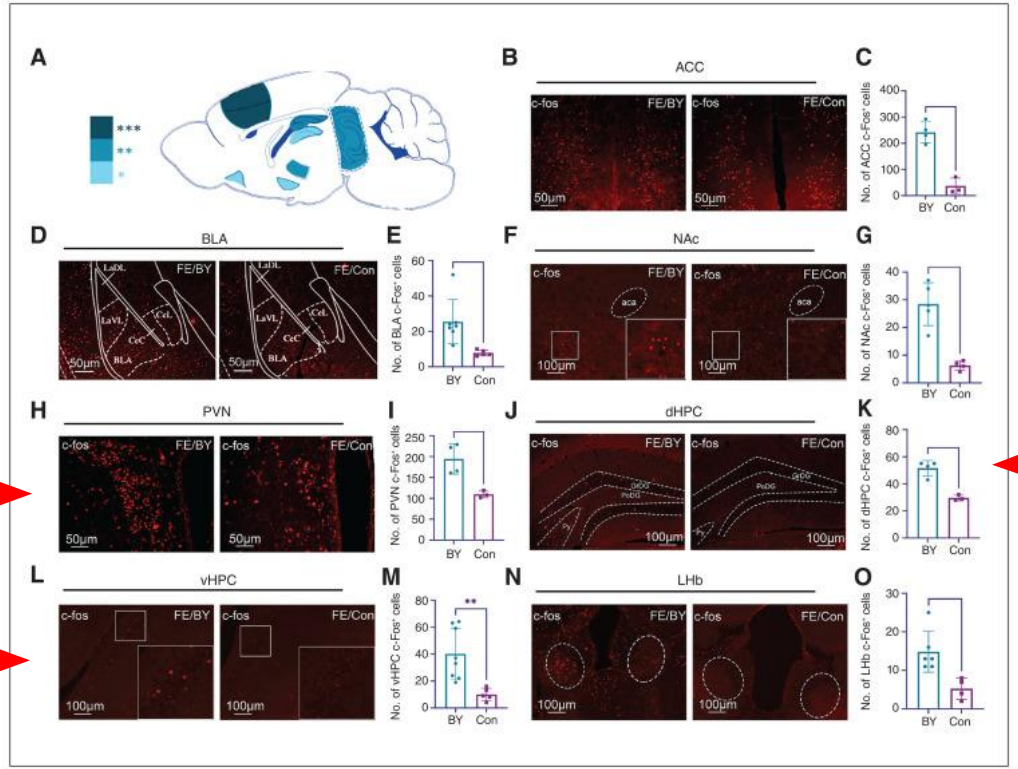
- No significant differences were found between their level of involvement with each type of empathy-like behaviours

Fig. 4

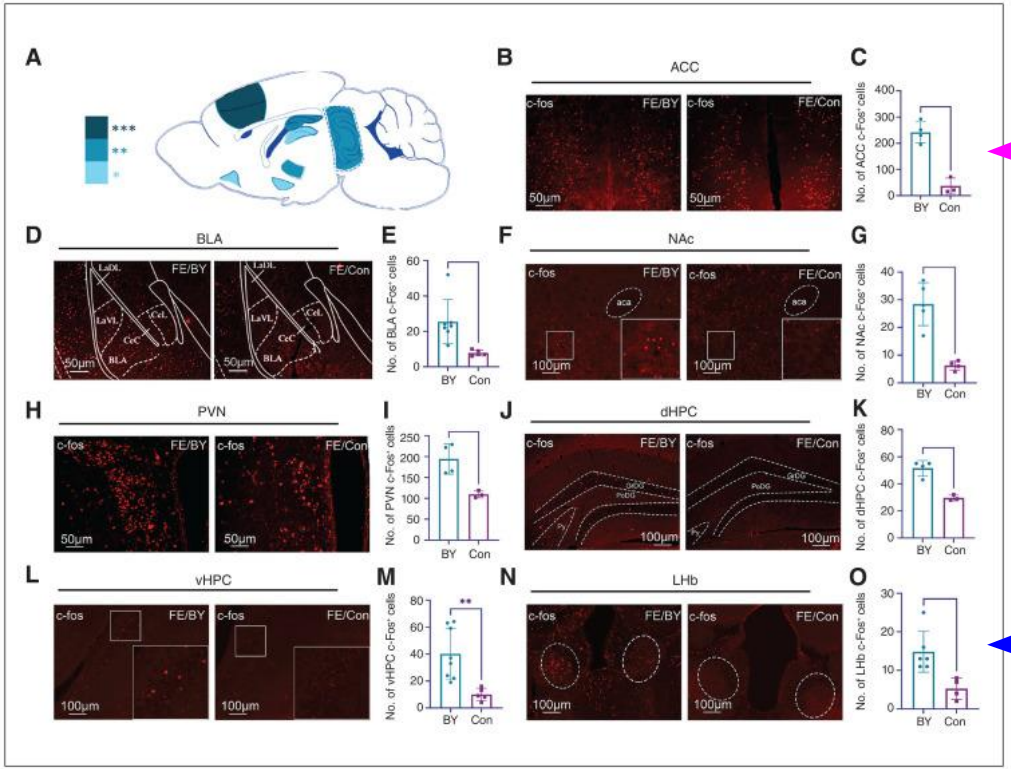
# Figure 4: Brain staining and c-Fos–positive cell counts in brain regions involved in the fear empathy in C57BL mice



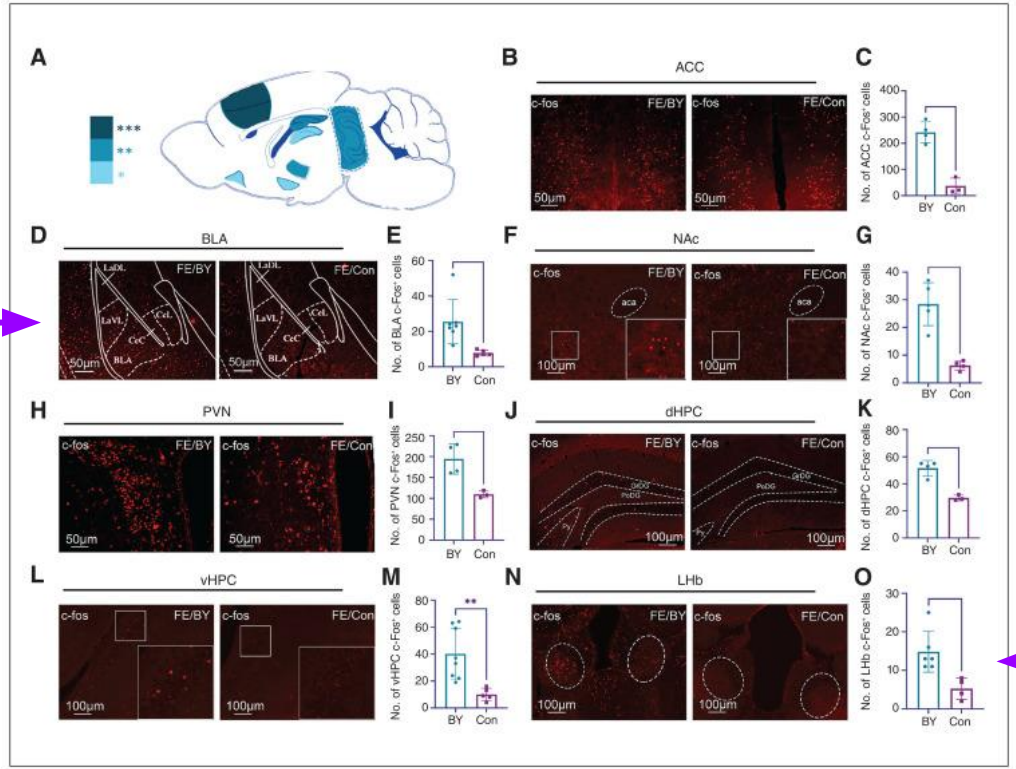
# Figure 4: Brain staining and c-Fos–positive cell counts in brain regions involved in the fear empathy in C57BL mice



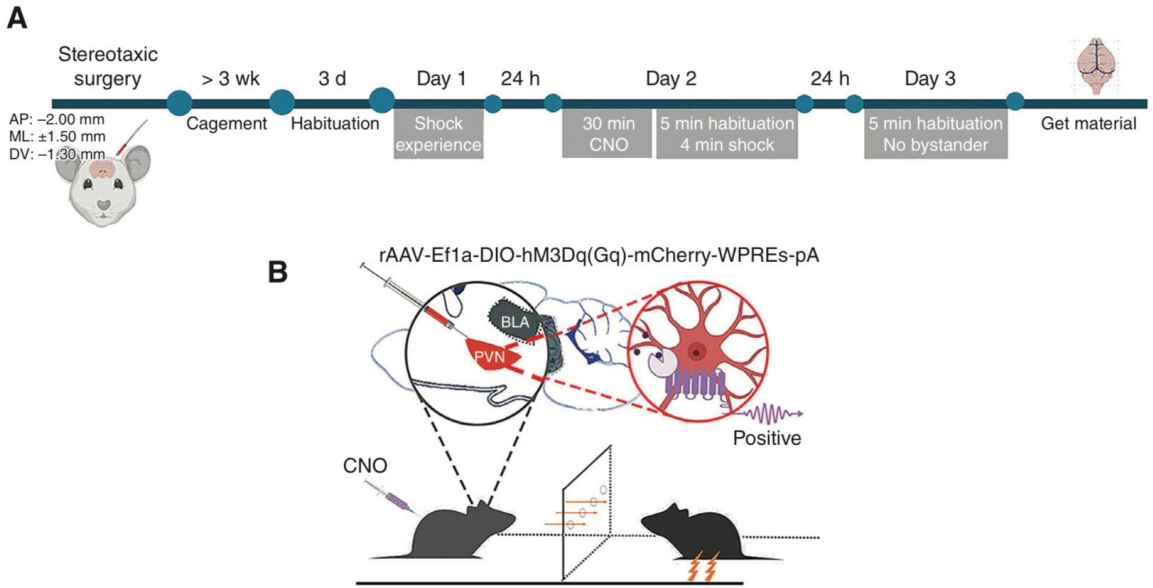
# Figure 4: Brain staining and c-Fos–positive cell counts in brain regions involved in the fear empathy in C57BL mice



# Figure 4: Brain staining and c-Fos–positive cell counts in brain regions involved in the fear empathy in C57BL mice



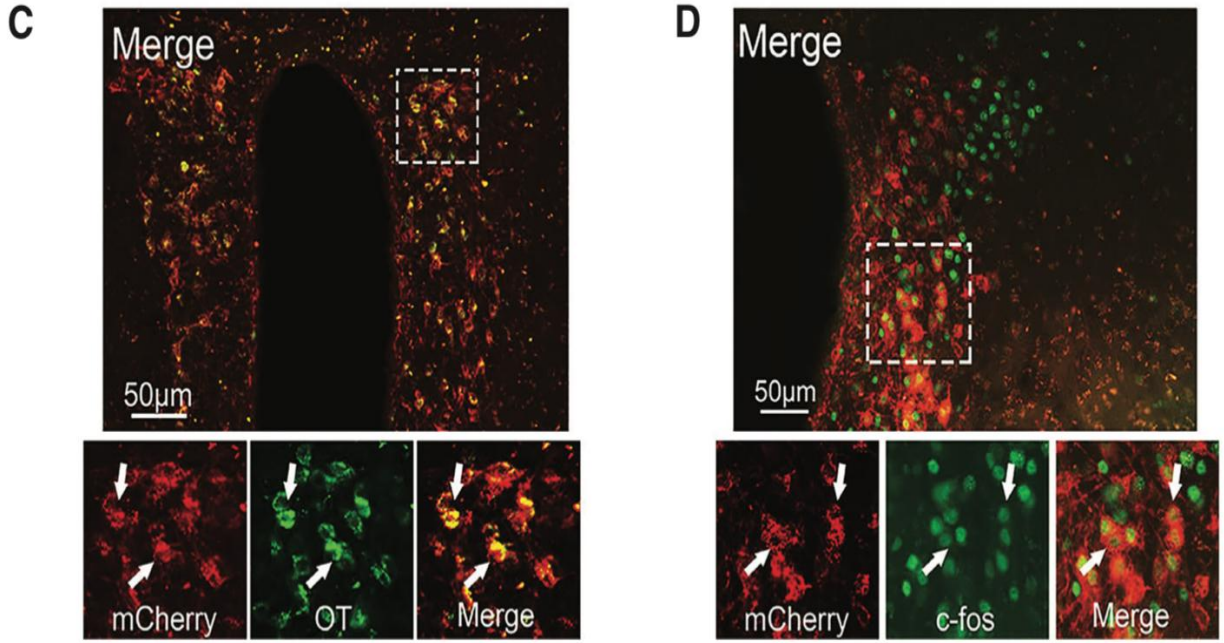
# Figure 8: Activating Oxytocin Neurons using Chemogenetics




**Key Takeaway**: Experimental timeline, indicating when the viral vectors were injected, when mice recovered, and when chemogenetic activation and behavioral testing occurred.

**Method** : virus carrying a Cre-dependent activating receptor (hM3Dq) was injected into the PVN of oxytocin-Cre mice, enabling selective activation of OT neurons with CNO.

# Figure 8: Usage of Chemogenetics

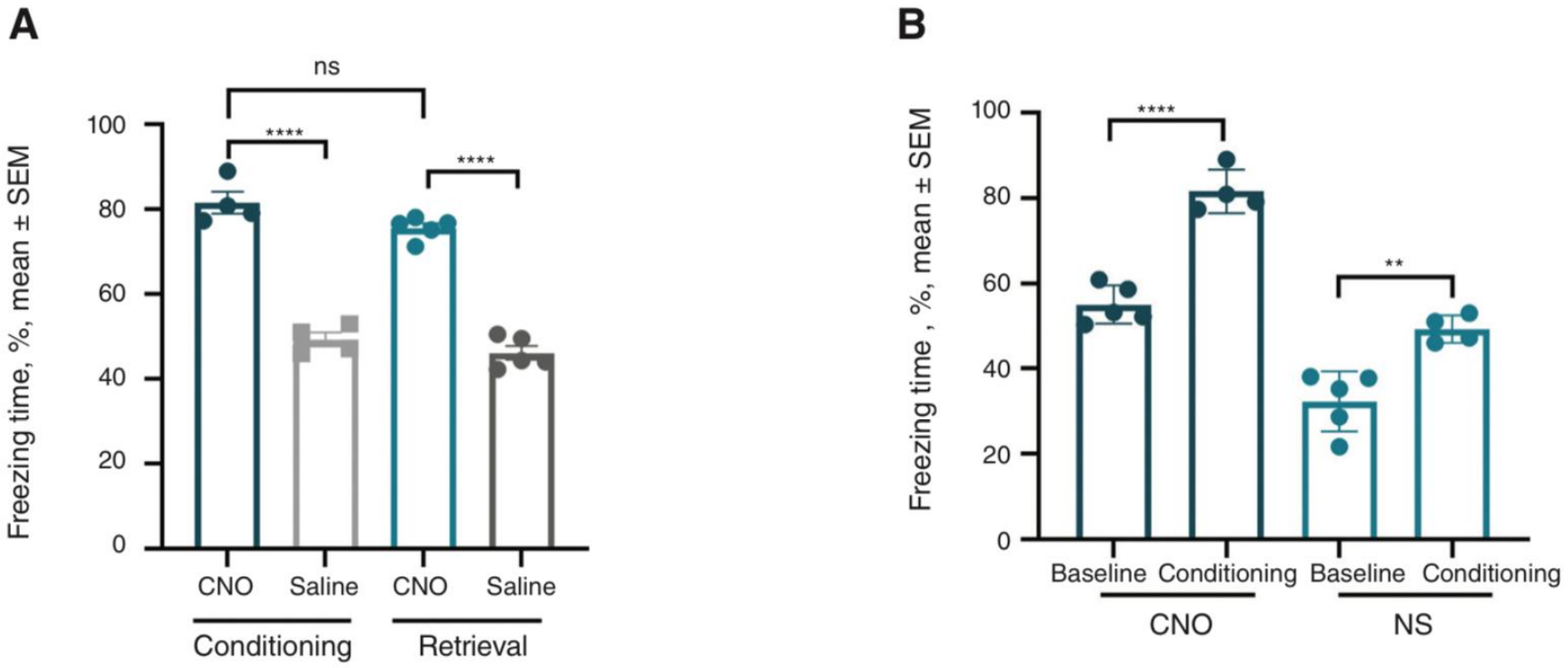



**Key Takeaway**:

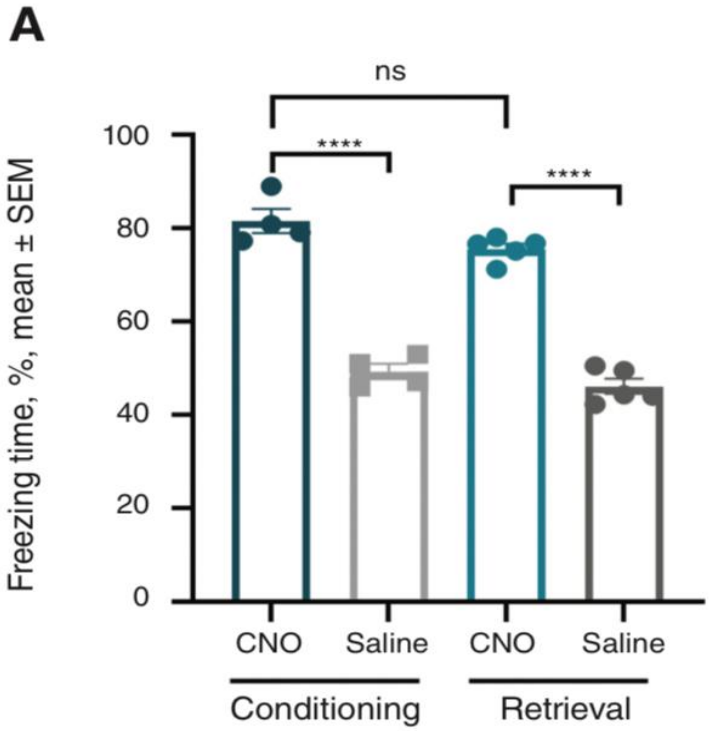
C → Nearly all neurons expressing the mCherry fluorescent marker also express oxytocin. Conformed that the chemogenetic activation system was selectively expressed in oxytocin-producing neurons within the PVN.

D → Functional activation of targeted neurons. Co-expression confirms that the chemogentic system effectively activated that oxytocin neurons in the PVN upon CNO administration.

# Figure 9: Panel A and B



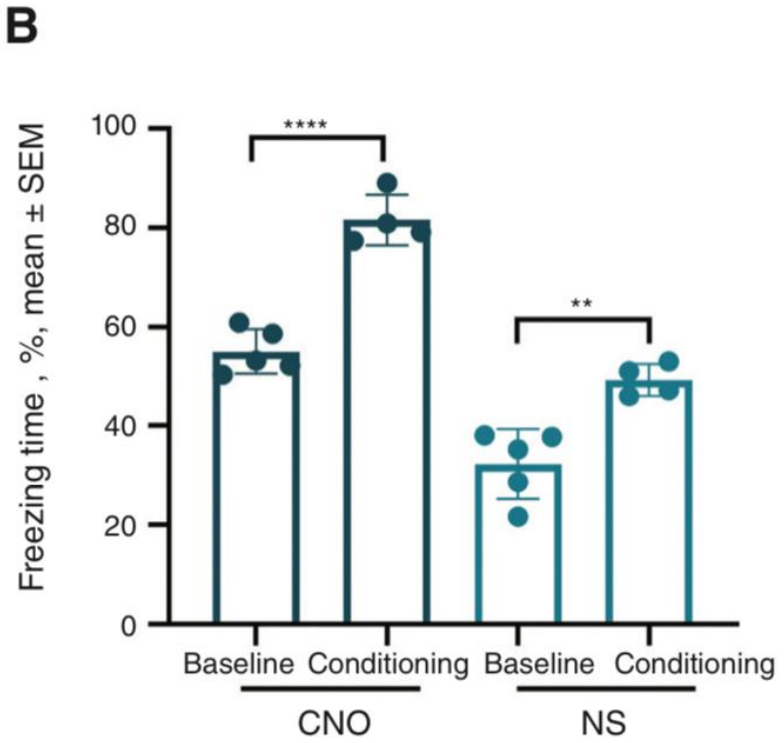
# Figure 9: Behavioral Effects of Oxytocin Neuron Activation




**Key Takeaway:** Mice with oxytocin neuron activation showed more freezing behavior during empathy-related fear tests.

- Enhanced emotional response to another's distress.
- **No significance** between conditioning and retrieval periods.

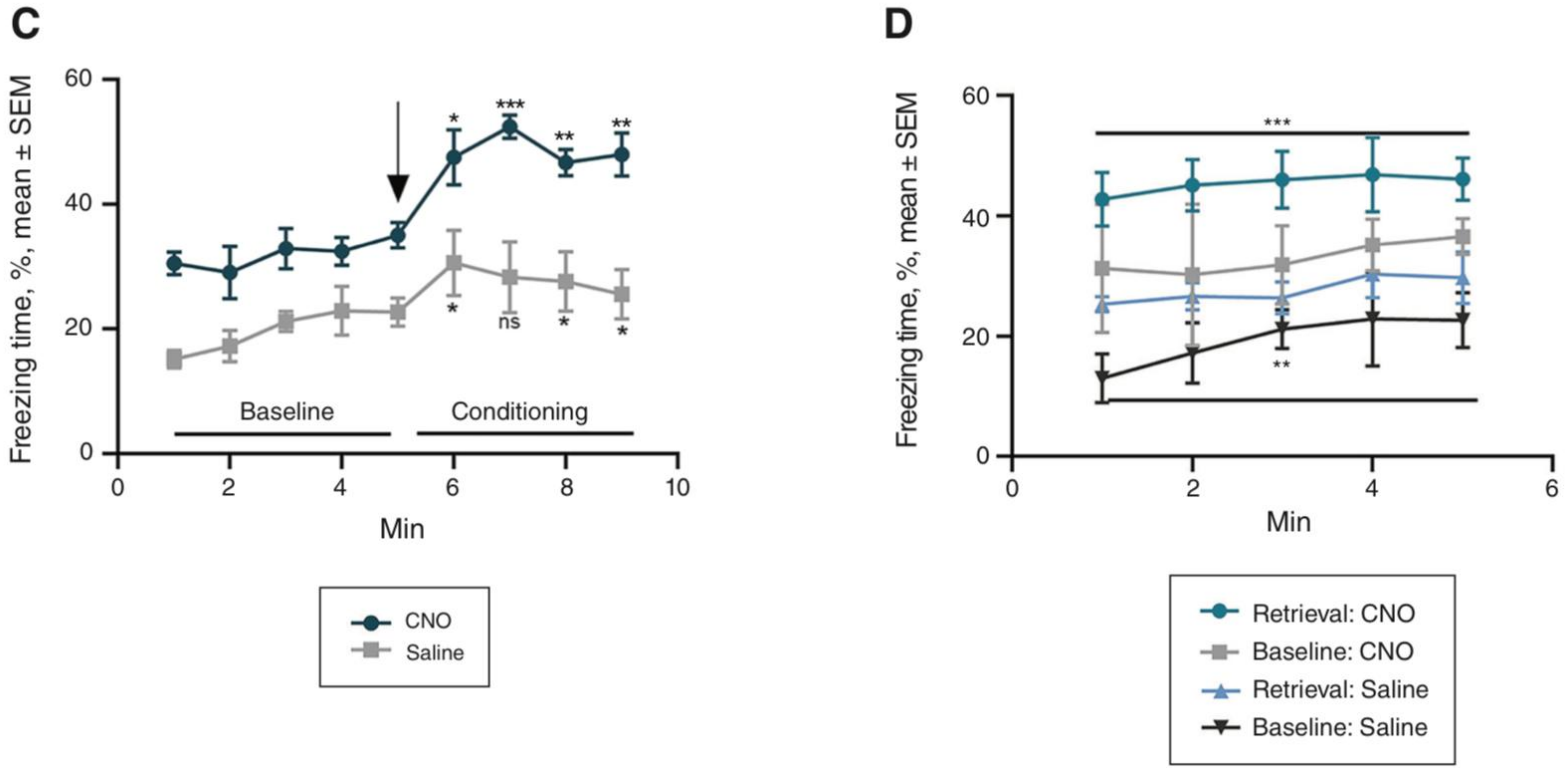
# Figure 9: Activation of PVN Oxytocin Neurons Enhances Empathy-Related Freezing Behaviors



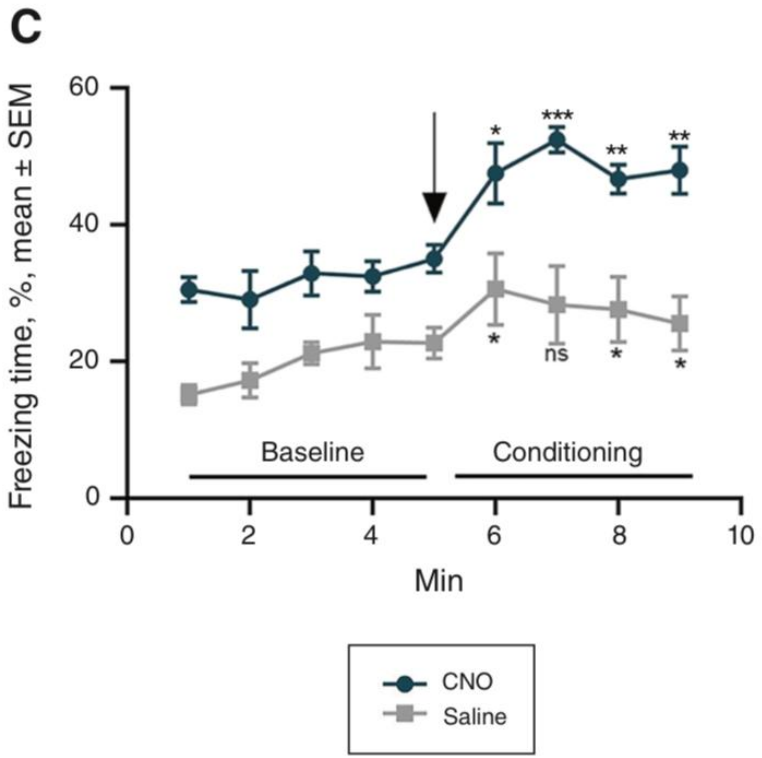

**Key Takeaway:** Increase in freezing behavior from baseline was significantly higher in CNO-treated mice.

- Oxytocin neuron activation amplifies fear empathy.
- CNO group more sensitive to fear emotion

# Figure 9: Panel C and D



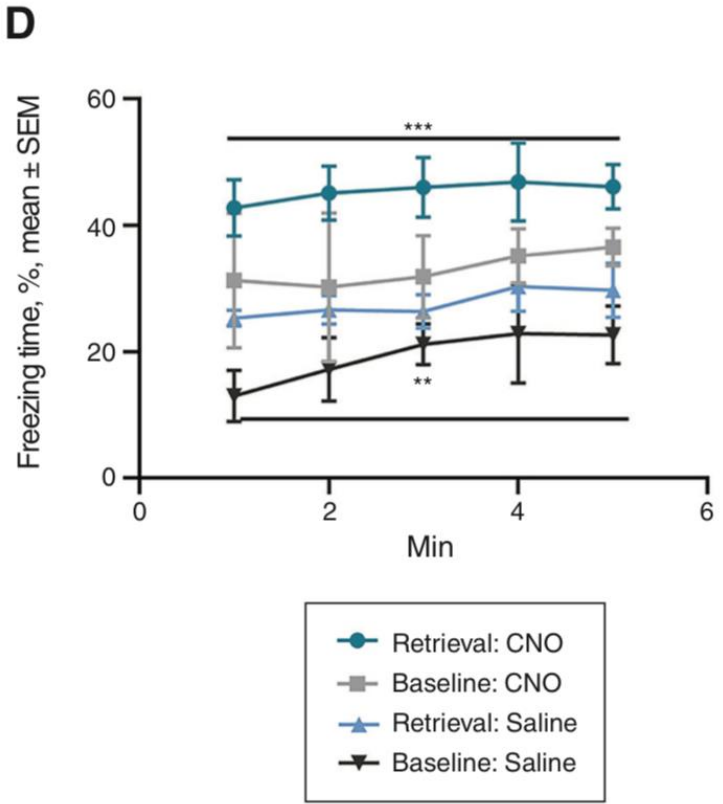
# Figure 9: Sustained Freezing Behavior during Fear Conditioning




**Key Takeaway:** Across the fear conditioning session (Day 2), CNO-treated mice maintained higher levels of freezing throughout conditioning.

- Suggesting a sustained effect of oxytocin activation on social fear learning.
- **Significant** increase in freezing time during fear conditioning period.

# Figure 9: Lasting Impact of Oxytocin Neuron Activation on Fear Memory Retrieval



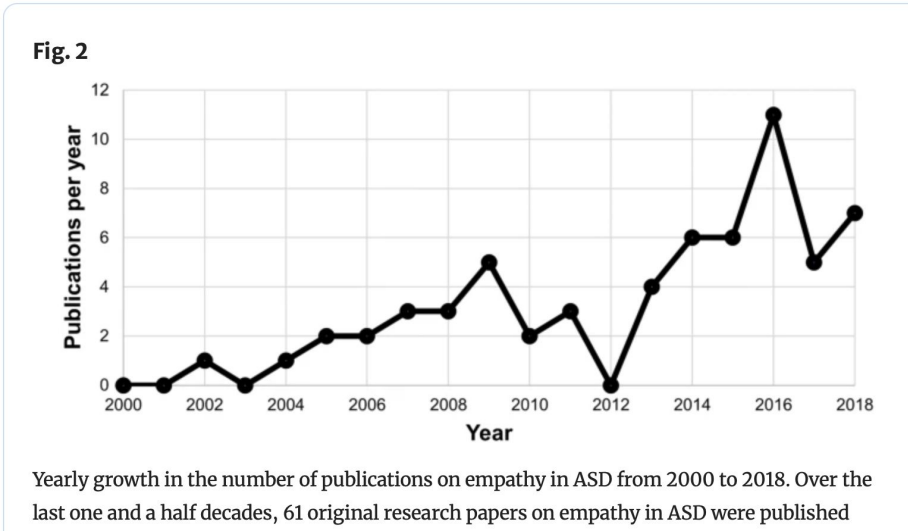

**Key Takeaway:** Following day (Day 3), even without further CNO, the CNO group continued to freeze more.

- Showing lasting effect on fear memory retrieval, likely influenced by prior oxytocin neuron activation.
- Freezing time of saline group gradually decreased while CNO group maintained high level.

# Summary

| | |
|---|---|
| Figure 1 | Mice show empathy like fear responses when they watch other mice in distress and have prior exposure to the fear box |
| Figure 2 | Bystander mice are more sensitive to pain after being exposed to a demonstrator mice being in pain |
| Figure 6 | Different brain regions were activated for Pain Empathy and Fear empathy models |
| Figure 4 | LHb and ACC are activated in both empathy-like responses, with LHb more involved in pain and ACC more involved in fear. BLA and NAc had more c-Fos–positive cells in the BY group for both empathy-like behaviors, this didn't differ between pain vs. fear. PVN, dorsal, and ventral hippocampus were selectively activated during empathy-like fear responses, with PVN showing the strongest activation. |
| Figure 8 | This figure confirms that the chemogenetic system effectively and selectively activates PVN oxytocin neurons, setting the stage for behavioral experiments. |
| Figure 9 | Activating PVN oxytocin neurons heightens both immediate and long-term empathy-like freezing behavior in response to social fear cues. |

# Further Evidence




Article: Harmsen, I.E. Empathy in Autism Spectrum Disorder. *J Autism Dev Disord* 49, 3939–3955 (2019). https://doi.org/10.1007/s10803-019-04087-w

Why this article?

- Shows how social behavior, emotional cognition, and empathy are all different in individuals with conditions like ASD
- The article we just presented showed how social pain and emotional cognition in mice ties closely with the concept of empathy.
- It is important that more research is done specifically for empathic and pain threshold in individuals with ASD since this article touched only a little on autistic individuals

# THANK YOU!
# Questions?